



PONTIFICIA UNIVERSIDAD CATÓLICA DE CHILE
IA LAB



IA Lab

Class 5 - Convolutional Neural Networks

IALAB UC

Computer Science Department, PUC

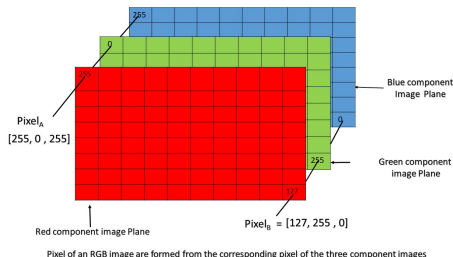
What do we want to solve?

Problems with *Unstructured Data*:
Images, text, audio, video.



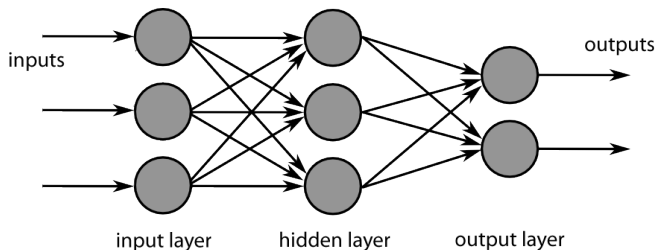
Unstructured data is just numbers

- **Images:** 3 color channels (RGB), intensity (0-255).¹
- **Audio:** frequency data with a time dimension. More on your Audio course!
- **Video:** images with a time dimension. More on your Video course!
- **Text:** Words can be turned to One-Hot vectors. More on your NLP course!

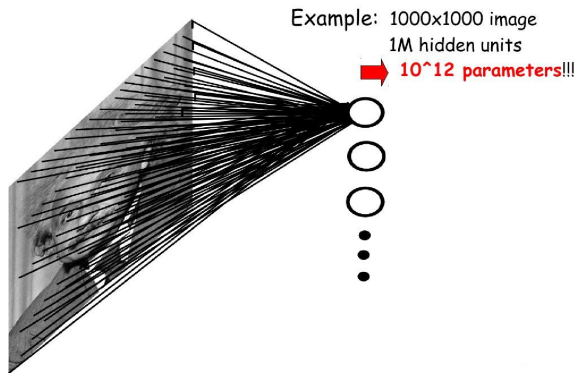


¹ Image taken from: <https://www.geeksforgeeks.org/matlab-rgb-image-representation/>

Traditional approach:
Fully connected feed-forward architecture.
(Remember your last course!)

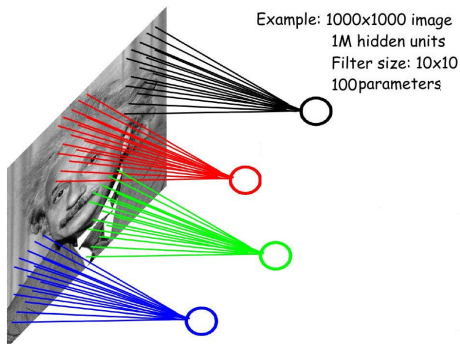


Supervised Neural Nets: Problems



- Huge number of parameters.
- It does not explicitly model local visual pattern (holistic approach).

Solution: Spatially Localized Neurons



- Spatially localized neurons provide a significant reduction in number of params. In the example, weights associated to each neuron drop from 10^6 to 10^2 .
- Furthermore, these neurons allow the net to capture local image patterns.

Problem

- A **local pattern** can potentially occur anywhere in the input.

What can we do about it?

- **Convolution** provides a mechanism to search for a given pattern (feature) by sliding a set of weights (**or filter**) through the spatial span of the input.
- Intuitively, the filter has a higher response (number) when it finds a matching pattern.
- Thus, we can think of a **filter** as detecting one pattern.
- Let's see an example!

Convolution: How does it work?

- A *convolution*² consists of elementwise multiplication of weights (**filter**) with input.
- We sum all products.
- Thus, result is a scalar (**neuron**).
- Then we repeat these operations by moving the filter across the image. In this case, by 1 box. We say the **stride** of the convolution is thus 1.
- The set of results (**feature map**) is the righthand matrix.
- This is a convolution in 2 **dimensions** because our input is in 2 dimensions.

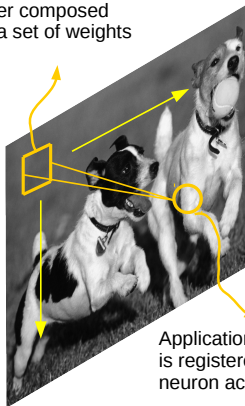
3	3	2	1	0
0	0	1	3	1
3 ₀	1 ₁	2 ₂	2	3
2 ₂	0 ₂	0 ₀	2	2
2 ₀	0 ₁	0 ₂	0	1

12	12	17
10	17	19
9	6	14

²Full Explanation: https://calvinfeng.gitbook.io/machine-learning-notebook/supervised-learning/convolutional-neural-network/convolution_operation

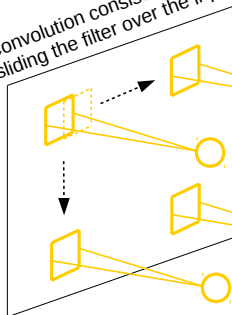
Convolutions: Overview

Filter composed by a set of weights



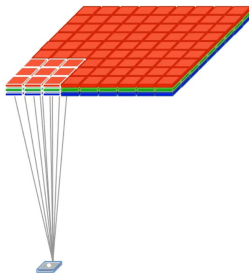
Application of filter is registered as a neuron activation

Convolution consists on sliding the filter over the input

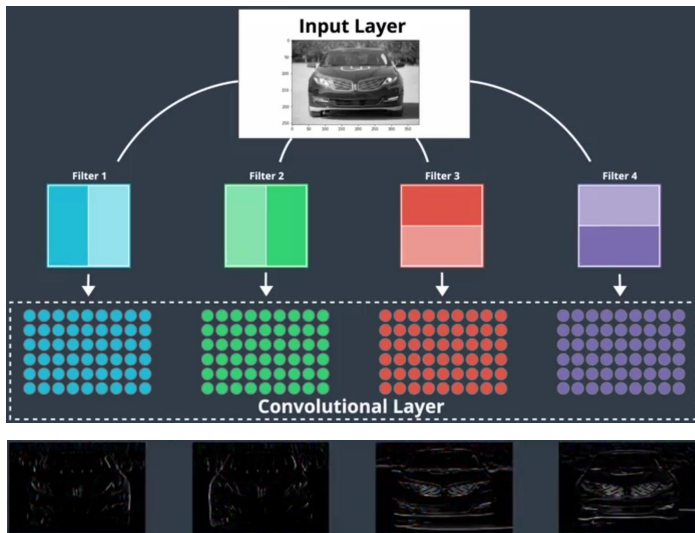


Set of all neurons generated by a given filter forms a feature map

Convolution Example in Motion!



Example: Convolving Filters Over an Input Image ³

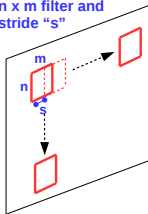


³Image from audacity.com

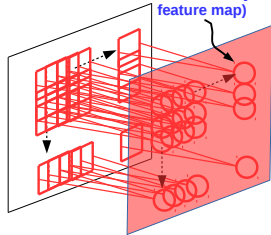
Convolution: Key Terms

- **Filter:** Set of values to be applied to input data. A matrix.
- **Neuron:** Output of the application of a filter. A number.
- **Stride:** How many elements to skip before applying filters again.
- **Feature Map:** The set of all output neurons for a specific filter.
- **Channels:** components of our data. Think of the RGB channels in images.
- **Dimension:** corresponds to the dimensionality of our data. For instance, images are 2D, videos 3D.

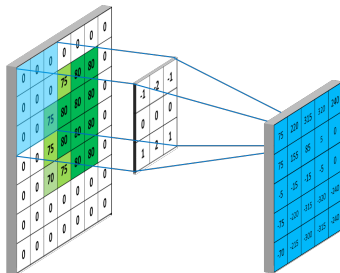
Convolution using a $n \times m$ filter and stride "s"



Neurons that compose a network layer (a.k.a feature map)



Exercise



- What is the input? How many dimensions (features) in the input?
- What is the filter? How many weights in the filter?
- What is the feature map? How many neurons in the feature map?

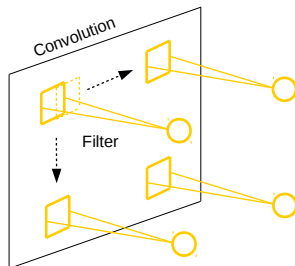
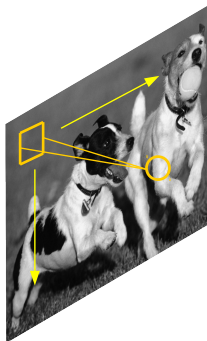
Convolutions: Advantages

- Output is **spatially localized** (i.e. we do not lose spatial information).
- **Reduced parameter size** since we are sharing weights with filters. Filters (i.e. weights) are independent from the location of the pattern in the input.
- Convolutions are cheap operations.
- In the old days, filters were handcrafted. But now, they can be **learned**!
- How? With **Convolutional Neural Networks**!

3	3	2	1	0
0	0	1	3	1
3 ₀	1 ₁	2 ₂	2	3
2 ₂	0 ₂	0 ₀	2	2
2 ₀	0 ₁	0 ₂	0	1

12	12	17
10	17	19
9	6	14

Convolutional Neural Networks (CNNs)

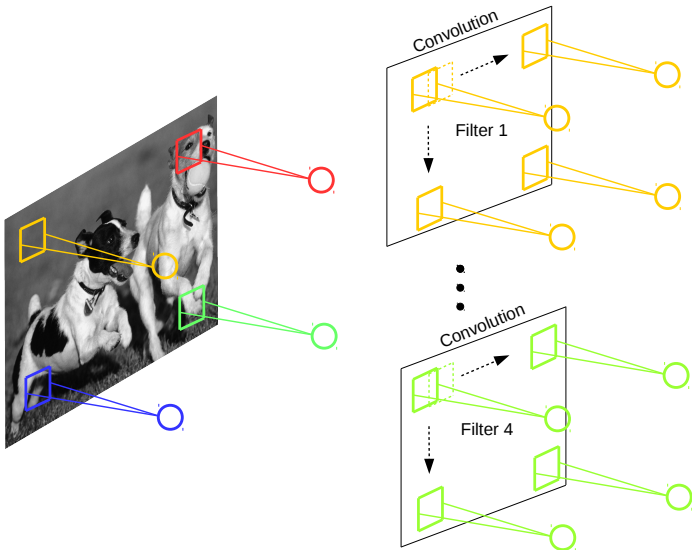


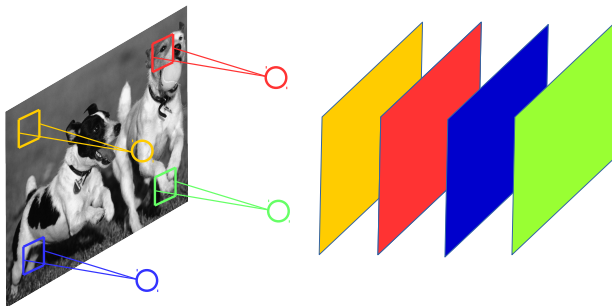
- Neural network that uses convolutions and learns the appropriate filters.
- Not only does it learn them but also learns a hierarchy of them.
- Let's see what this means!

**Detecting one pattern is not enough,
what can we do?**



Let's use several filters i.e. a Filter Bank

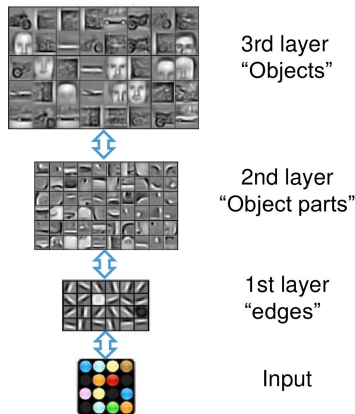




- We can use (**learn**) several filters.
- The convolution process generates an activation map for each filter, which is given by a set of neurons that **share weights**.
- By sharing weights, the model presents an improved mechanism to deal with the spatial variability of local patterns (how?).
- Each set of shared weights allows the net to learn a filter specifically tuned to detect a particular local pattern (filter banks).
- With N filters we can think of the output as a new image, but with N different channels instead of the usual 3!

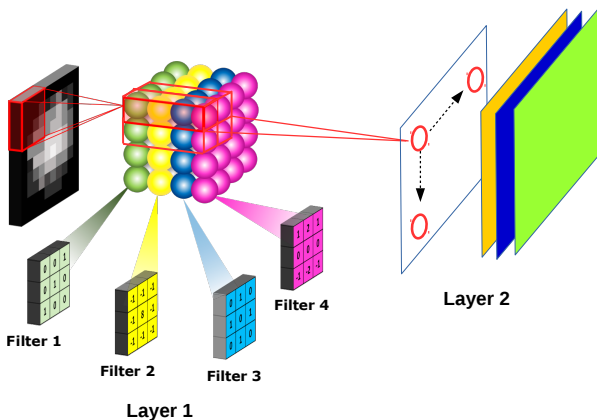
What do we mean by **deep**?
A hierarchical compositional
architecture.

Hierarchical Compositional Architecture

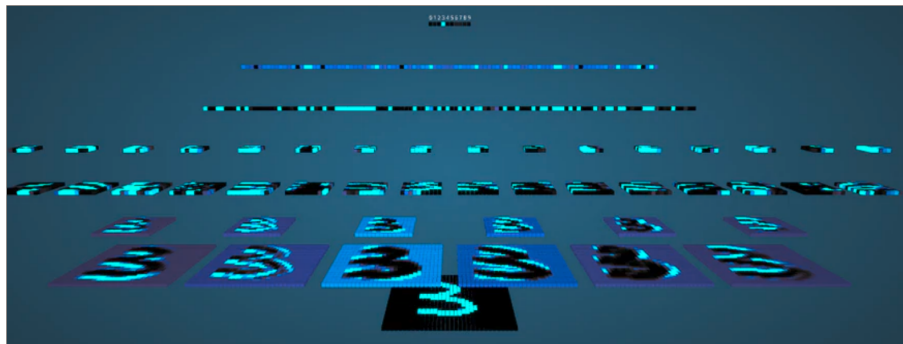


- Deep neural networks exploit the property that many natural signals form compositional hierarchies.
- Higher-level features are obtained by compositions of lower-levels.

A Hierarchy of Compositional Network Layers



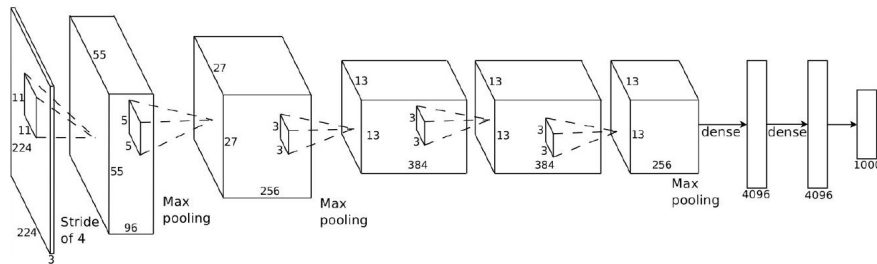
Example: Convolving Filters Over an Input Image



- Adam Harley's CNN visualization tool.
- www.cs.cmu.edu/~aharley/vis/conv/flat.html.

Going Deeper: Stack Several Layers of CNNs

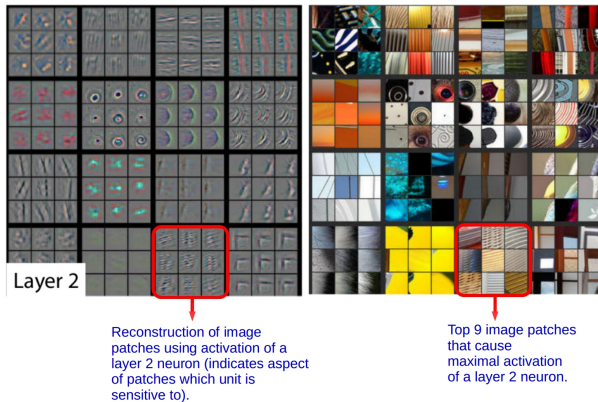
Example: Meet AlexNet (Krizhevsky et al., 2012)



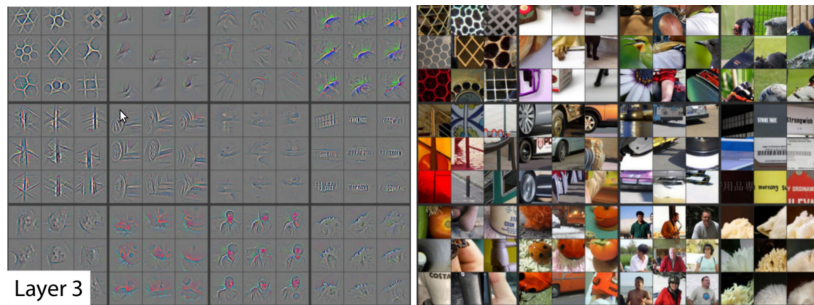
- In AlexNet⁴, the role of each convolutional layer is to detect local conjunctions of features from the previous layer.

⁴ <https://papers.nips.cc/paper/4824-imagenet-classification-with-deep-convolutional-neural-networks.pdf>

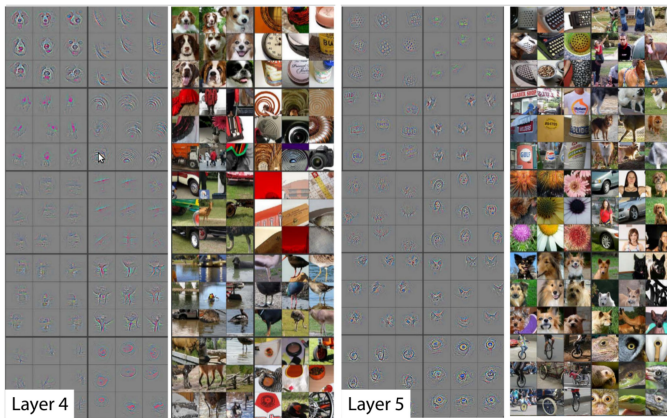
Ex. Visualization of Neuron Activations



Ex. Visualization of Neuron Activations

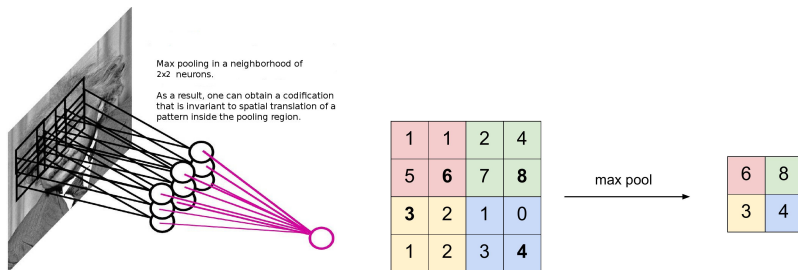


Ex. Visualization of Neuron Activations



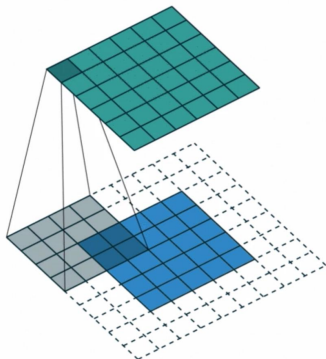
A few extra things to make CNNs
work...

Max Pooling Operation to Avoid Spatial Variability



- Max-pooling provides a mechanism to coarse-graining the spatial position of each feature for the next layers.
- These allows the net to incrementally reduce the dimensionality of the representation and to provide certain degree of invariance to small shifts and distortions of a given pattern.
- In other words, the pooling allows representations to vary very little when elements in the previous layer present some variations in local position and appearance.

Zero Padding Operator



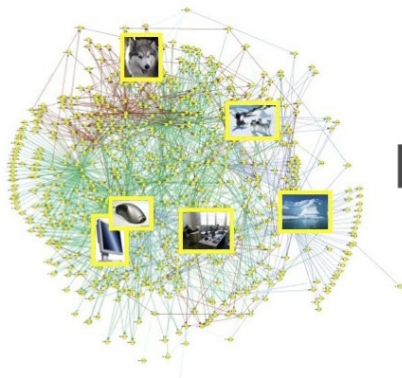
- By padding with zeros the borders of the images, it is possible to apply the convolution operator in these areas.
- Thus, we do not lose information from the borders which might be relevant.

Case Example

ImageNet Classification with Deep Convolutional Neural Networks

A. Krizhevsky, I. Sutskever, and G. E. Hinton (NIPS-2012)

- 15M images.
- 22K categories.
- Images collected from Web.
- Human labelers (Amazon's Mechanical Turk crowd-sourcing).



IMAGENET

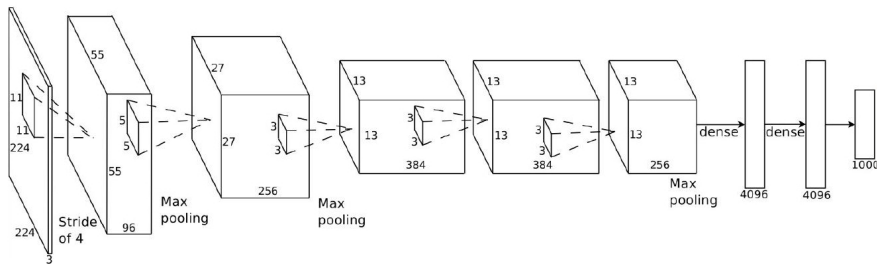
ILSVRC-2010

- 1K categories.
- 1.2M training images (1000 per category).
- 50K validation images.
- 150K testing images.
- RGB images.
- Variable-resolution, but in this application images are scaled to 256x256 pixels.

Classification goals

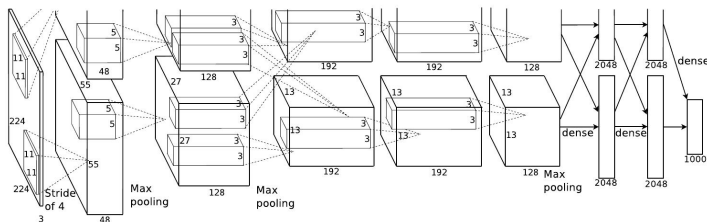
- Make 1 guess about the label (Top-1 error).
- make 5 guesses about the label (Top-5 error).

AlexNet Architecture:



Pay attention! We will be working with this architecture in the first laboratory!

Architecture: Trained on 2 GPUs



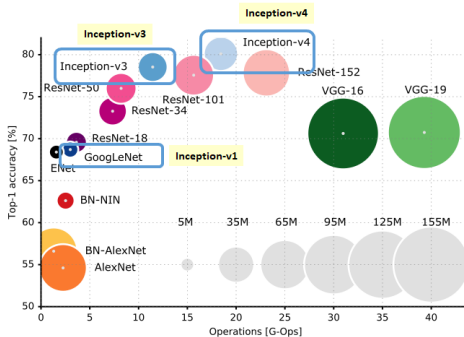
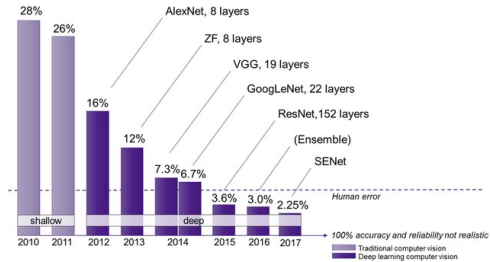
- Trained with stochastic gradient descent on two NVIDIA GTX 580 3GB GPUs for about a week.

Results on ILSVRC-2012 competition

- Top-5 error rate: 15.3%.
- 2nd best team: 26.2% error.
- Human error: approx 8%.

Network size

- Aprox. 650,000 neurons.
- 60,000,000 parameters.

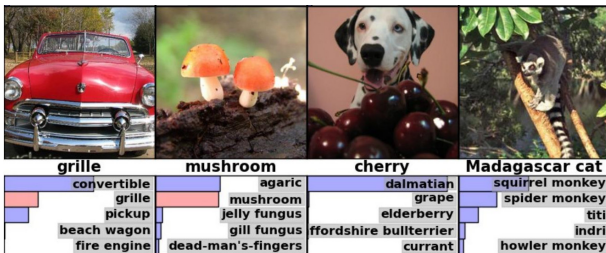


Results on ILSVRC

Results on ILSVRC-2012 competition:

- Top-5 error rate: 15.3%.
- 2nd best team: 26.2% error.
- Human error approx. 8%.

Take these numbers with care, as an example, the following cases correspond to wrong classifications:





lens cap

reflex camera
Polaroid camera
pencil sharpener
switch
combination lock



abacus

abacus
typewriter keyboard
space bar
computer keyboard
accordion



slug

slug
zucchini
ground beetle
common newt
water snake



hen

hen
cock
cocker spaniel
partridge
English setter



tiger

tiger
tiger cat
tabby
boxer
Saint Bernard



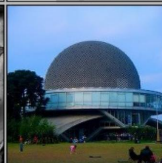
chambered nautilus

lampshade
throne
goblet
table lamp
hamper



tape player

cellular telephone
slot
reflex camera
dial telephone
iPod



planetarium

planetarium
dome
mosque
radio telescope
steel arch bridge



koala



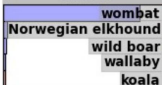
tiger



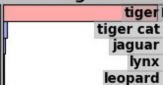
European fire salamander



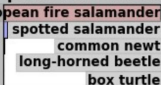
loggerhead



seat belt



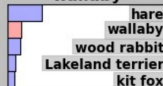
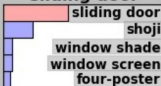
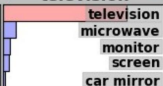
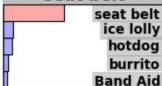
television

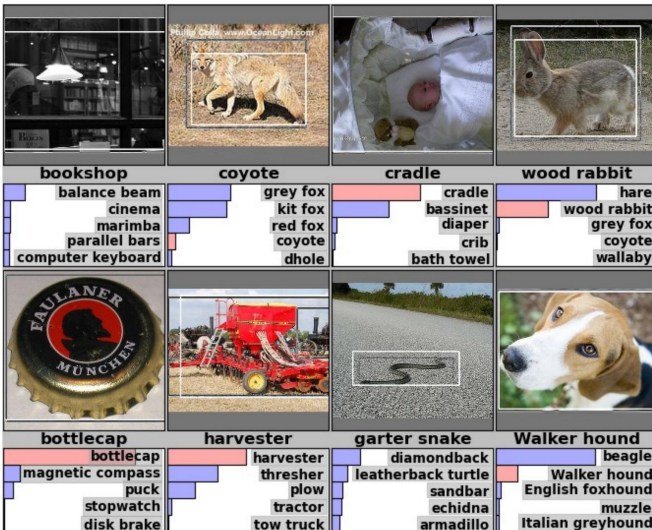


sliding door



wallaby





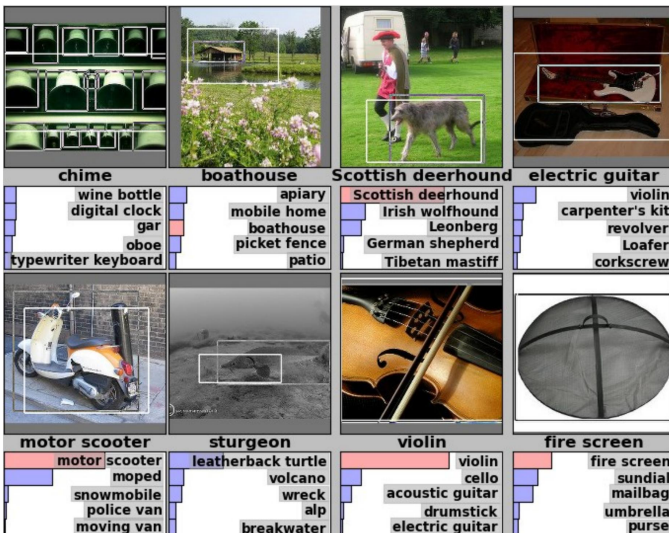


Image Retrieval

First column contains query images from ILSVRC-2010 test set, remaining columns contain retrieved images from training set.

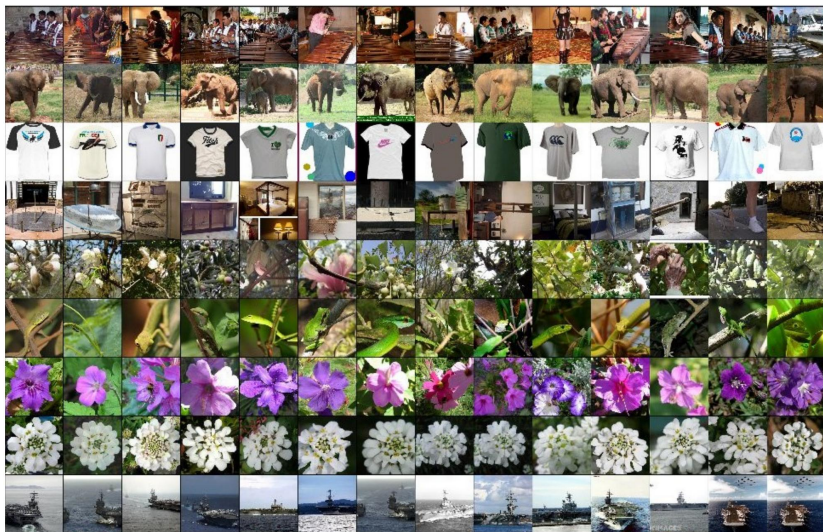
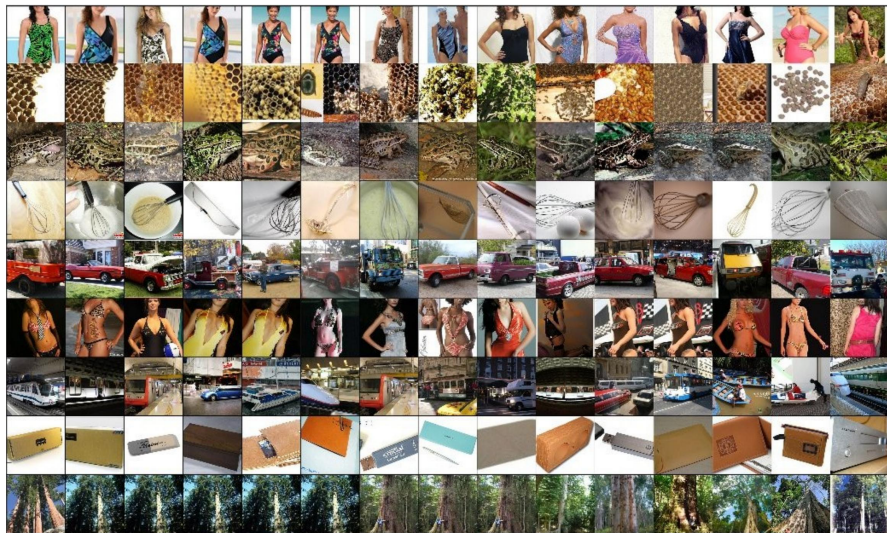


Image Retrieval



Some Classes in ImageNet Are Very Specific and Similar



(a) Siberian husky



(b) Eskimo dog

Today, several closely related but deeper CNN architectures, have reduced the previous error rate. As an example, the work by K. Simonyan and A. Zisserman achieves a Top-5 error rate of 6.8% (deep: 16-19 weight layers).⁵

⁵ K. Simonyan and A. Zisserman. *Very Deep Convolutional Networks for Large-Scale Image Recognition*. International Conference on Learning Representations, 2015.

While the previous CNN was trained on ImageNet, a nice property is that the **learned features can be useful to other visual applications**.

- For example, most deep learning frameworks offer different models pretrained on ImageNet.
- In Lab 2, we will be working with an AlexNet model pretrained on ImageNet.

- Convolutional Neural Networks do not work with Image data only.
- We can make them work with text and audio data as well!
- Let's see some examples!

Case Example

Text Mining and Convolutional Neural Networks

Kim et al., 2014.

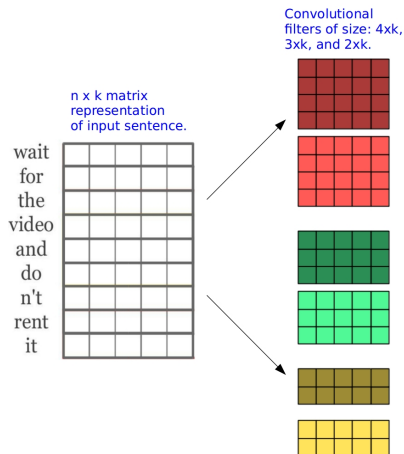
A Simple CNN Model for Text Mining (Kim et al., 2014)

$n \times k$ matrix representation
of input sentence.

wait					
for					
the					
video					
and					
do					
n't					
rent					
it					

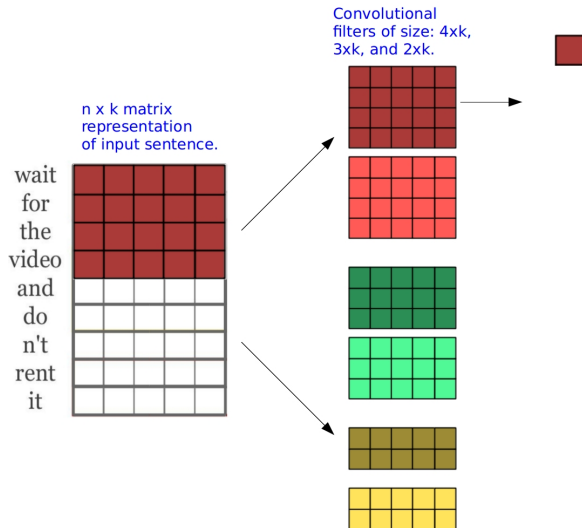
- Input is a sentence matrix, where each row corresponds to the coding of a sentence word using Word2Vec.
- Therefore, for a sentence of n words and a coding using k dimensions, the input matrix has size $n \times k$.
- Kim et al. use a 300D Word2Vec model trained on 100 billion words of Google News ($k=300$).

A Simple CNN Model for Text Mining (Kim et al., 2014)

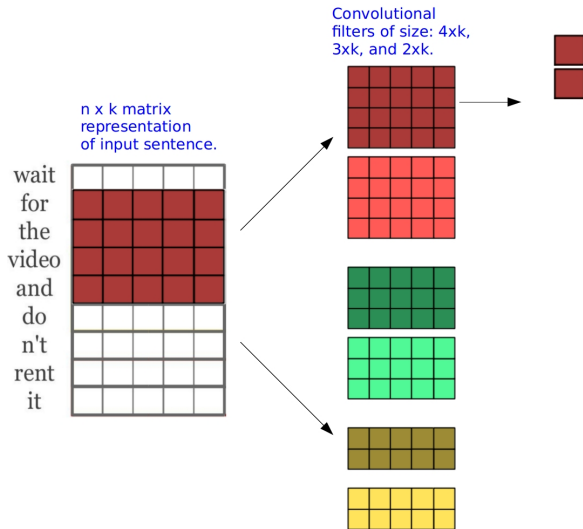


- As in AlexNet, a key step of CNN's training is learning weights of convolutional filters.

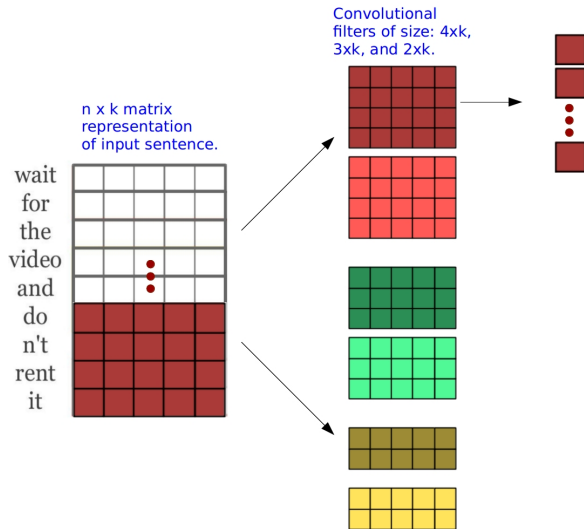
Convolution



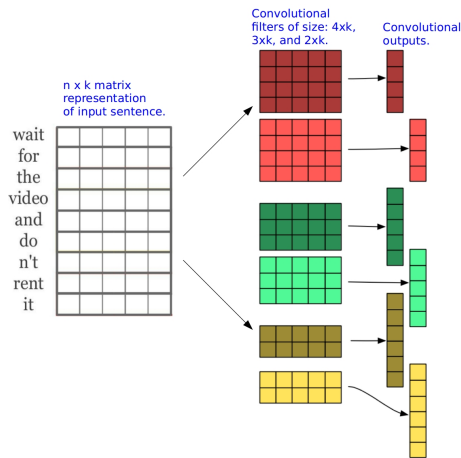
Convolution



Convolution

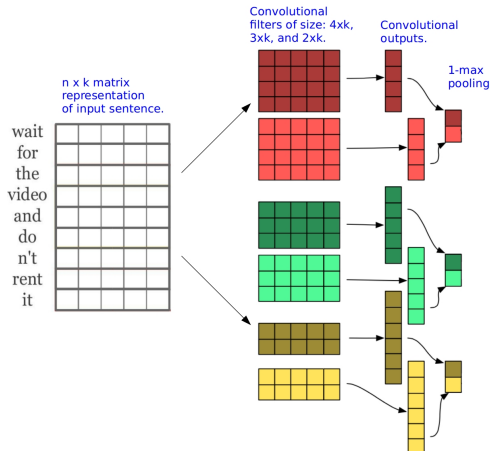


A Simple CNN Model for Text Mining (Kim et al., 2014)



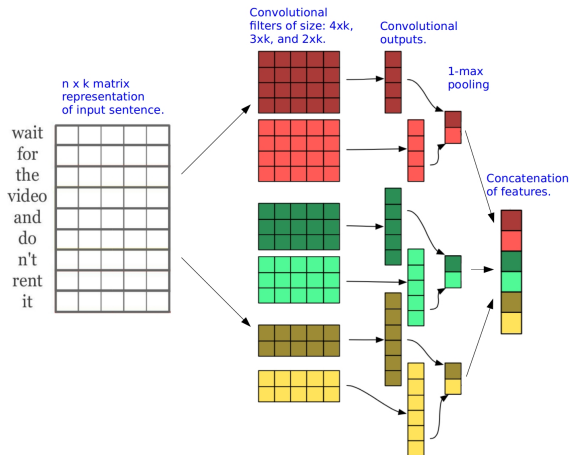
- Kim et al. use window filter sizes of 3,4,5 words (number of rows).
- Each filter size generates 100 feature maps.
- How many types of features do we learn with this filter architecture?

A Simple CNN Model for Text Mining (Kim et al., 2014)



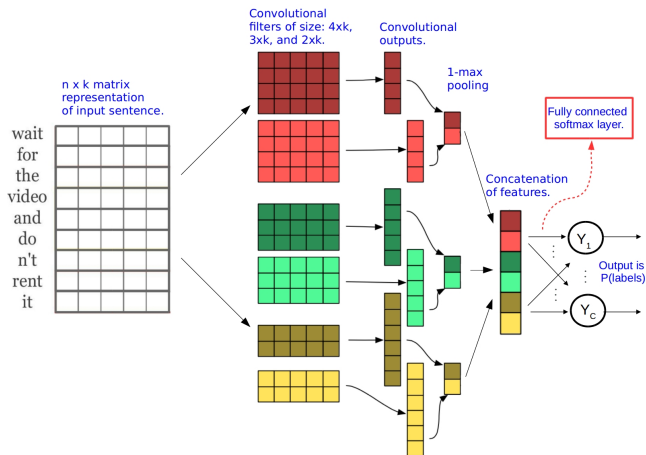
- Outputs from each filter are 1-max pooled (winner-takes-all).
- This pooling scheme naturally deals with variable sentence lengths.
- How does this pooling affect the sentence model? (Hint: think about word order in input sentence).

A Simple CNN Model for Text Mining (Kim et al., 2014)



- Outputs from filters are concatenated to form the feature vector that will feed the final classification step.

A Simple CNN Model for Text Mining (Kim et al., 2014)



- Complete architecture including a final fully connected layer with soft-max output.
- Soft-max units provide a probability estimate over target classes.

Benchmarks

- **MR:** Movie reviews with one sentence per review. Classification involves detecting positive/negative reviews (Pang and Lee, 2005).³
- **SST-1:** Stanford Sentiment Treebank—an extension of MR but with train/dev/test splits provided and fine-grained labels (very positive, positive, neutral, negative, very negative), re-labeled by Socher et al. (2013).⁴
- **SST-2:** Same as SST-1 but with neutral reviews removed and binary labels.
- **Subj:** Subjectivity dataset where the task is to classify a sentence as being subjective or objective (Pang and Lee, 2004).
- **TREC:** TREC question dataset—task involves classifying a question into 6 question types (whether the question is about person, location, numeric information, etc.) (Li and Roth, 2002).⁵
- **CR:** Customer reviews of various products (cameras, MP3s etc.). Task is to predict positive/negative reviews (Hu and Liu, 2004).⁶
- **MPQA:** Opinion polarity detection subtask of the MPQA dataset (Wiebe et al., 2005).⁷

Results

Model	MR	SST-1	SST-2	Subj	TREC	CR	MPQA
CNN-rand	76.1	45.0	82.7	89.6	91.2	79.8	83.4
CNN-static	81.0	45.5	86.8	93.0	92.8	84.7	89.6
CNN-non-static	81.5	48.0	87.2	93.4	93.6	84.3	89.5
CNN-multichannel	81.1	47.4	88.1	93.2	92.2	85.0	89.4
RAE (Socher et al., 2011)	77.7	43.2	82.4	—	—	—	86.4
MV-RNN (Socher et al., 2012)	79.0	44.4	82.9	—	—	—	—
RNTN (Socher et al., 2013)	—	45.7	85.4	—	—	—	—
DCNN (Kalchbrenner et al., 2014)	—	48.5	86.8	—	93.0	—	—
Paragraph-Vec (Le and Mikolov, 2014)	—	48.7	87.8	—	—	—	—
CCAE (Hermann and Blunsom, 2013)	77.8	—	—	—	—	—	87.2
Sent-Parser (Dong et al., 2014)	79.5	—	—	—	—	—	86.3
NBSVM (Wang and Manning, 2012)	79.4	—	—	93.2	—	81.8	86.3
MNB (Wang and Manning, 2012)	79.0	—	—	93.6	—	80.0	86.3
G-Dropout (Wang and Manning, 2013)	79.0	—	—	93.4	—	82.1	86.1
F-Dropout (Wang and Manning, 2013)	79.1	—	—	93.6	—	81.9	86.3
Tree-CRF (Nakagawa et al., 2010)	77.3	—	—	—	—	81.4	86.1
CRF-PR (Yang and Cardie, 2014)	—	—	—	—	—	82.7	—
SVM _S (Silva et al., 2011)	—	—	—	—	95.0	—	—

CNN-rand: initial word vectors are random.

CNN-static: initial word vector given by Word2Vec coding.

CNN-non-static: initial word vector given by Word2Vec are fine-tuned for each task.

CNN-multichannel: contain two sets of word vectors, one is kept fixed, other one is fine-tuned.

Results

	Most Similar Words for	
	Static Channel	Non-static Channel
bad	<i>good</i> <i>terrible</i> <i>horrible</i> <i>lousy</i>	<i>terrible</i> <i>horrible</i> <i>lousy</i> <i>stupid</i>
good	<i>great</i> <i>bad</i> <i>terrific</i> <i>decent</i>	<i>nice</i> <i>decent</i> <i>solid</i> <i>terrific</i>
n't	<i>os</i> <i>ca</i> <i>ireland</i> <i>wo</i>	<i>not</i> <i>never</i> <i>nothing</i> <i>neither</i>
!	<i>2,500</i> <i>entire</i> <i>jez</i> <i>changer</i>	<i>2,500</i> <i>lush</i> <i>beautiful</i> <i>terrific</i>
,	<i>decasia</i> <i>abysmally</i> <i>demise</i> <i>valiant</i>	<i>but</i> <i>dragon</i> <i>a</i> <i>and</i>

Based on cosine similarity for vectors in the static channel and fine-tuned vectors in the non-static channel. Results using the multichannel model and the SST-2 dataset.

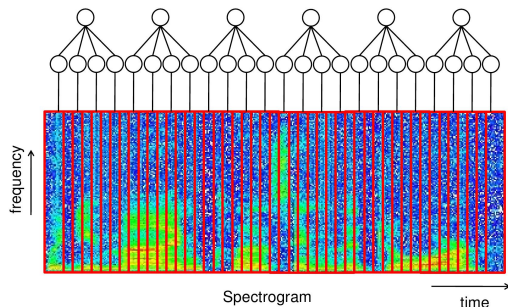
Application

Audio Classification and Convolutional Neural Networks

Hershey et al., 2017⁶

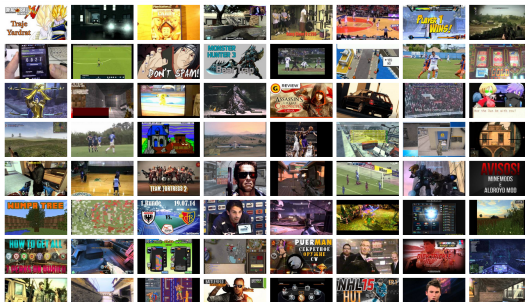
⁶CNN Architectures for Large Scale Audio Classification <https://arxiv.org/pdf/1609.09430v2.pdf>

Large Scale Audio Classification (Hershey et al., 2017)



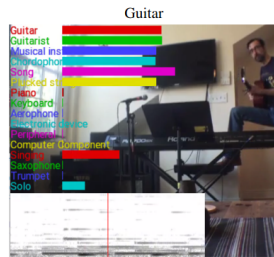
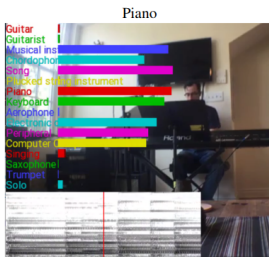
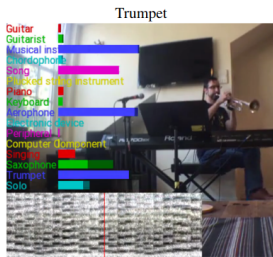
- They turn audio waveforms into a Mel Spectrogram.
- Mel Spectrogram has two relevant dimensions: frequency and time.
- We've turned an audio signal into an image!
- They divide audio from videos into constant length chunks.
- They classify according to each chunk, then average results across all chunks to produce final result.

Large Scale Audio Classification (Hershey et al., 2017)



- They use regular CNN architectures for images on Audio.
- AlexNet and others such as ResNet, VGG, Inception (we'll see these in class 3, don't worry!)
- They train on the audio of the YouTube-100M video dataset. (100MM videos!)
- Dataset has thousands of classes, quite challenging.

Results



Architectures	Steps	Time	AUC	d-prime	mAP
Fully Connected	5M	35h	0.851	1.471	0.058
AlexNet	5M	82h	0.894	1.764	0.115
VGG	5M	184h	0.911	1.909	0.161
Inception V3	5M	137h	0.918	1.969	0.181
ResNet-50	5M	119h	0.916	1.952	0.182
ResNet-50	17M	356h	0.926	2.041	0.212